AGRIAID+ : AI-POWERED SMART FARMING & CLIMATE RESILIENCE PLATFORM

GROUP NO. 1
SHARAYU MAHAJAN (22107051)
RISHI MANE (22107063)
KALPANA MOHANTY (22107059)
MONISH MUDALIAR (22107027)

PROJECT GUIDE
DR. VEENA TRIVEDI

01 PROBLEM STATMENT

- Farmers lack access to integrated, data-driven tools for crop selection, fertilizer usage, and market decisions, leading to low productivity and inefficiencies.
- Existing solutions are fragmented, non-personalized, and often not available in regional languages.
- There is a need for a unified AI-powered platform to provide real-time insights, personalized guidance, and better market access.

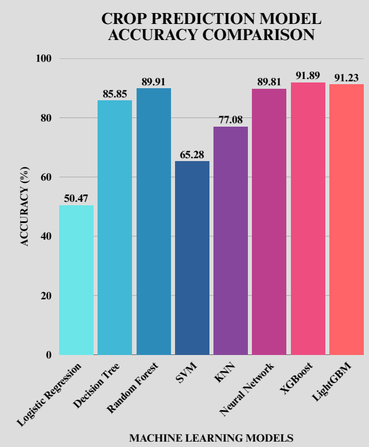
02 OBJECTIVE

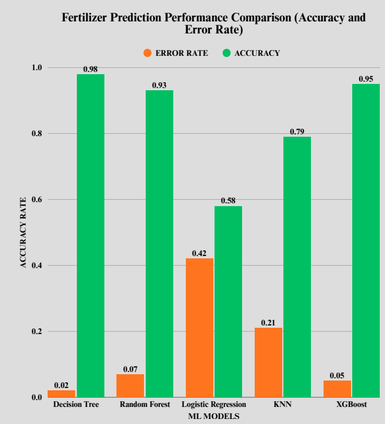
- Develop a personalized crop & fertilizer recommendation system using ML models like XGBoost and LightGBM
- Build a multilingual AI chatbot for farmer assistance in regional languages
- Integrate weather forecasting & market price analytics for smarter crop planning
- Create a farmer-centric digital marketplace for direct trading of produce and equipment

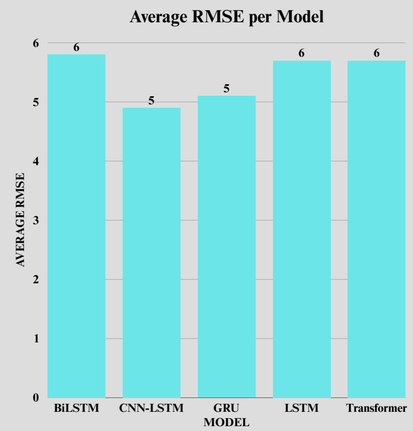
03 TECHNOLOGY STACK

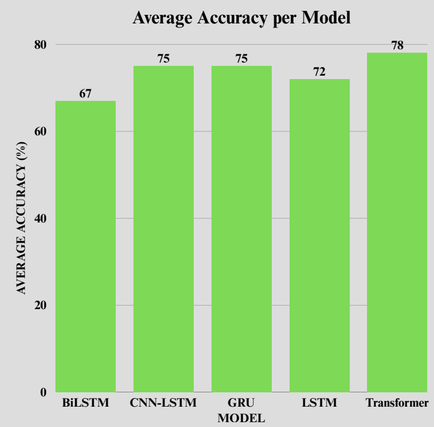
- Programming: Python
- ML Models: XGBoost, LightGBM, Scikit-learn
- Data Processing: Pandas, NumPy
- Frontend: HTML, CSS
- Backend: Flask
- Database: MongoDB
- APIs: Open Weather API
- Chatbot/NLP: Rag Based using LLM
- Others: PyEphem, Matplotlib

04 RESULT

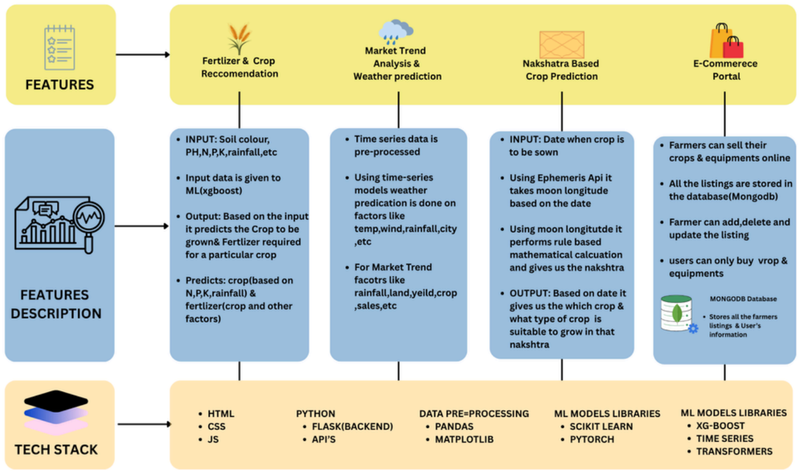








05 ARCHITECTURE DIAGRAM



06 RESEARCH GAP LIMITATIONS

- Limited language support excludes many farmers using local dialects
- Lack of trust (no ratings/feedback) in platforms like eNAM
- Small/noisy datasets reduce prediction accuracy
- Weather and climate factors often ignored
- High computational cost limits ML accessibility
- Absence of long-term climate data for planning